

FETH model editing

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I) Preparation

1) Getting the necessary files

This modding method assumes that you'll take an existing g1m as a basis and modify/replace its submeshes. In this tutorial, we want to get Sakura into the game. Anette seems like a good candidate as a basis (same size) so I got Anette's head g1m.

	1.exe	2/24/2020 3:38 PM	Application	42 KB
	2.exe	2/15/2020 4:46 PM	Application	2,480 KB
	3.exe	2/15/2020 4:46 PM	Application	2,492 KB
	4.exe	2/24/2020 3:14 PM	Application	22 KB
	Anette.g1m	11/23/2019 3:24 PM	G1M File	1,734 KB

2) Installing the blender plugins

In Blender go to edit > Preferences and install the “blender_3dmigoto.py” and “blender_FETH.py” plugins. The first one will be for model import/export, the second one will be used for weight painting and skeleton previewing.

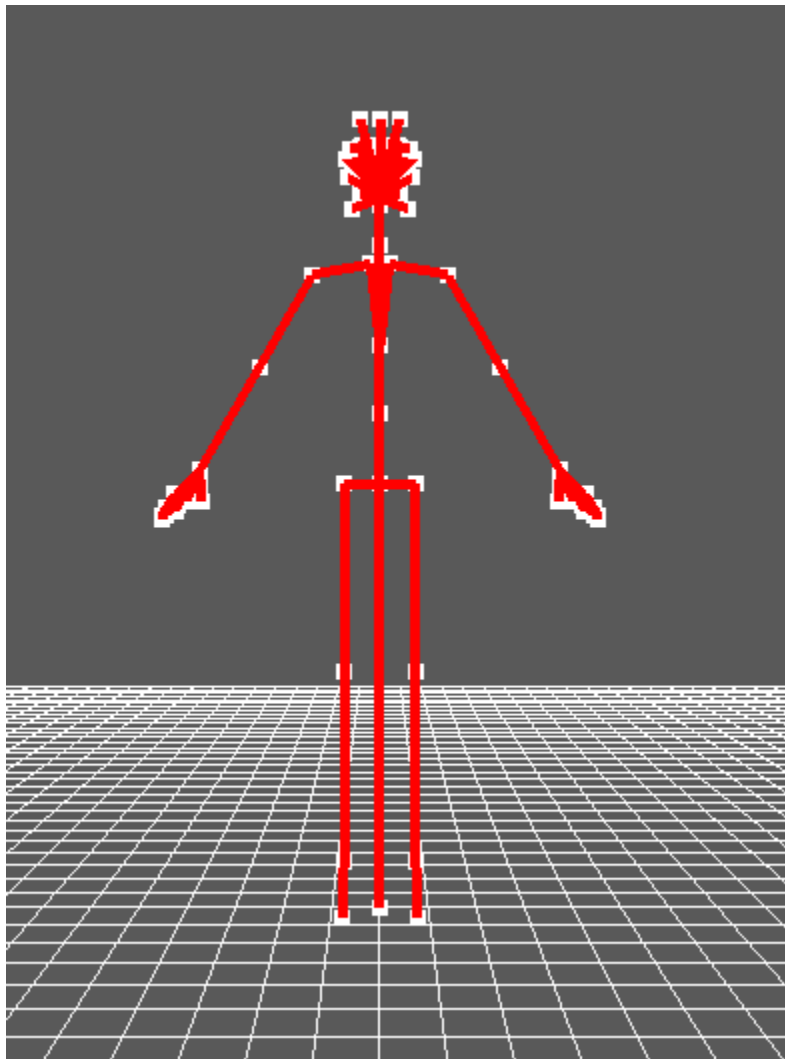
Put the “noesis_FETH_skel.py” file in the Noesis python folder. It will be used to get the skeleton from the base g1m and preview the weight painting in Blender. **If you use my fmt_g1m.py plugin for g1m viewing, make sure to get rid of it first or it will be used by Noesis instead!**

3) Getting the skeleton and the ib/vb

Here's what we have so far

 Scripts	3/20/2020 10:06 PM	File folder	
 1.exe	2/24/2020 3:38 PM	Application	42 KB
 2.exe	2/15/2020 4:46 PM	Application	2,480 KB
 3.exe	2/15/2020 4:46 PM	Application	2,492 KB
 4.exe	2/24/2020 3:14 PM	Application	22 KB
 Anette.g1m	11/23/2019 3:24 PM	G1M File	1,734 KB
 Sakura.fbx	3/22/2020 3:57 PM	FBX File	1,090 KB

The scripts folder won't be used anymore once you installed the addons. Anette.g1m is the base g1m file, Sakura.fbx is the model I want to put into the game. Open Anette.g1m with the Noesis plugin you've just installed.



If you don't see anything press the skeleton icon in Noesis. Export that file as an fbx, I named it Anette_skel.fbx

Here's what we have now:

	Scripts	3/20/2020 10:06 PM	File folder	
	1.exe	2/24/2020 3:38 PM	Application	42 KB
	2.exe	2/15/2020 4:46 PM	Application	2,480 KB
	3.exe	2/15/2020 4:46 PM	Application	2,492 KB
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	Anette.g1m	11/23/2019 3:24 PM	G1M File	956 KB
	Anette_skel.fbx	3/22/2020 4:10 PM	FBX File	152 KB
	boneMap.bonebind	3/22/2020 4:07 PM	BONEBIND File	3 KB
	Sakura.fbx	3/22/2020 3:57 PM	FBX File	1,090 KB

The bonebind file holds the several maps to the local bone indices used by each submesh, we'll use that in Blender later.

Time to generate the submeshes file, drag and drop the "Anette.g1m" to "1.exe"

	Scripts	3/20/2020 10:06 PM	File folder	
	1.exe	2/24/2020 3:38 PM	Application	42 KB
	2.exe	2/15/2020 4:46 PM	Application	2,480 KB
	3.exe	2/15/2020 4:46 PM	Application	2,492 KB
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	FE_Anette.g1m	3/22/2020 4:13 PM	G1M File	963 KB
	Sakura.fbx	3/22/2020 3:57 PM	FBX File	1,090 KB

It will produce a new g1m named "FE_original_name.g1m". Drag and drop this new file to "2.exe".

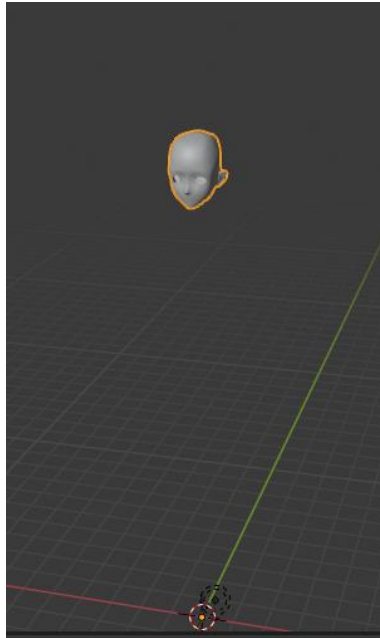
	FE_Anette	3/22/2020 4:14 PM	File folder	
	Scripts	3/20/2020 10:06 PM	File folder	
	1.exe	2/24/2020 3:38 PM	Application	42 KB
	2.exe	2/15/2020 4:46 PM	Application	2,480 KB
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It will output some folder with the same name and all the submeshes files in a format readable by the 3dmigoto plugin. Everything is set now, time to launch Blender.

II) Modifying the files in Blender

1) Importing the files

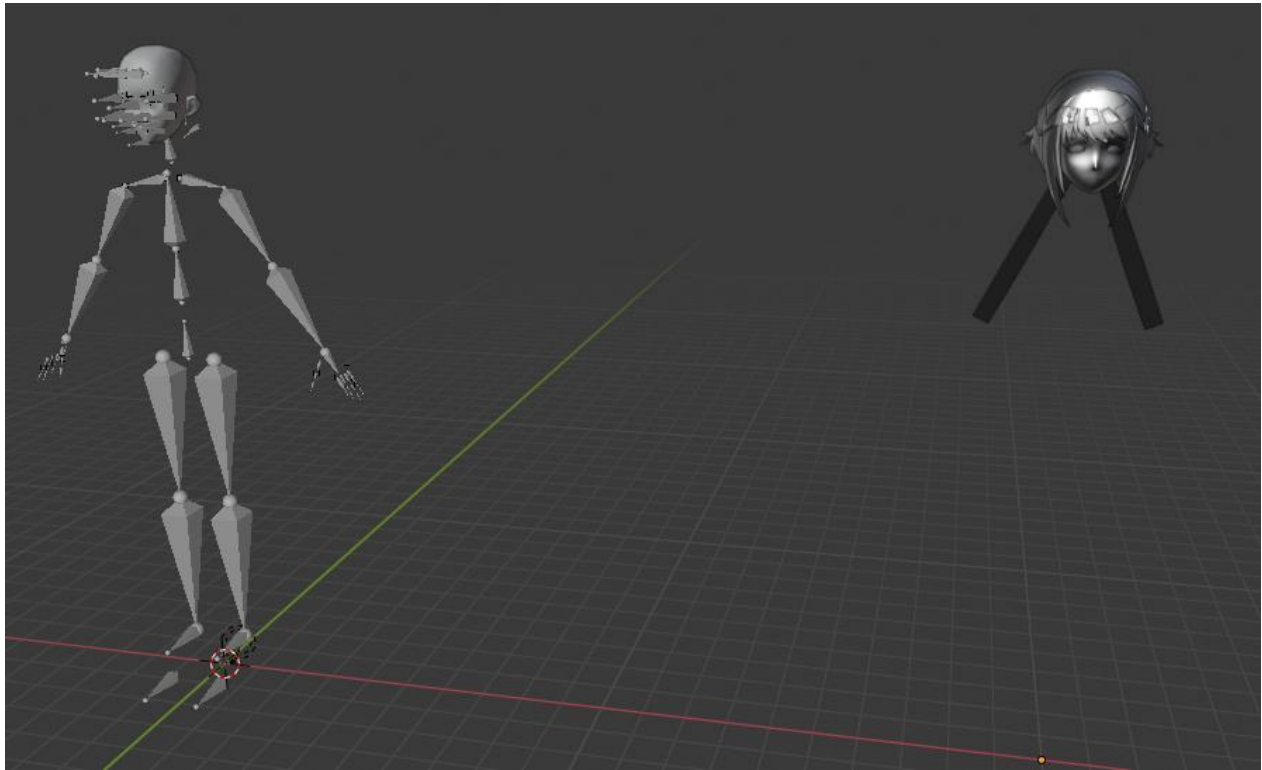
To import the different submeshes simply go to file> import > 3D migoto raw buffers. In this example I only want to replace Anette's face by Sakura's, and only on the highest LOD submesh. That mesh is 0.vb, so select the 0.ib/0.vb and click on import. You can also import several submeshes at the same time.



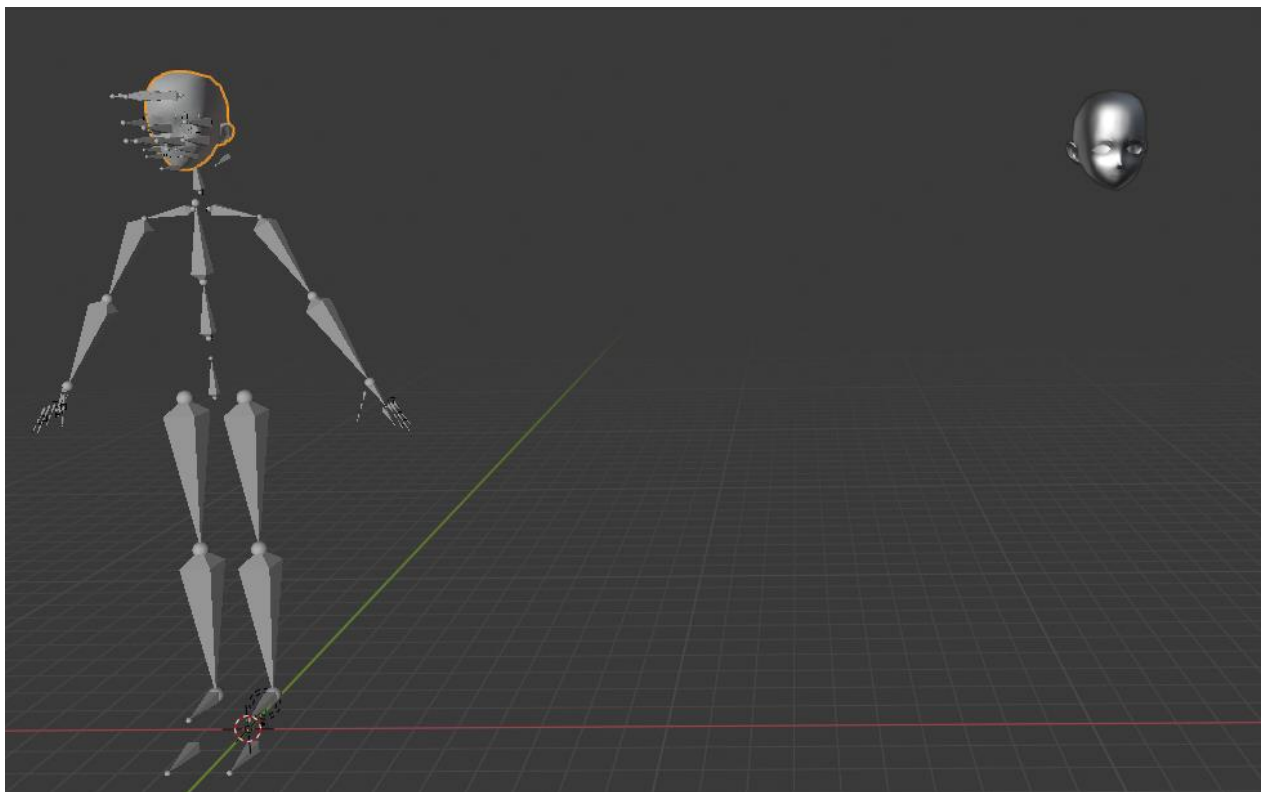
Let's import the skeleton too: file > import > fbx > Anette_skel.fbx. I recommend using the automatic rotation option so that rotations aren't as broken as without the option.



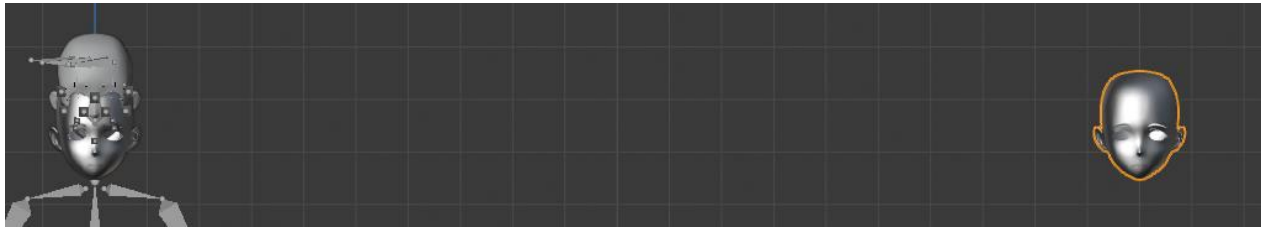
Finally let's import Sakura's mesh, I deleted her armature and driver meshes and applied a 200 units translation for better visibility.



I'll only work on her face here, so I delete the other parts.



Duplicate the 0.vb mesh (Anette's face. Name one of them "base_0" and the other "ref_0". Select base_0 then Shift select Sakura's head mesh. Press CTRL+L and choose "Object Data":



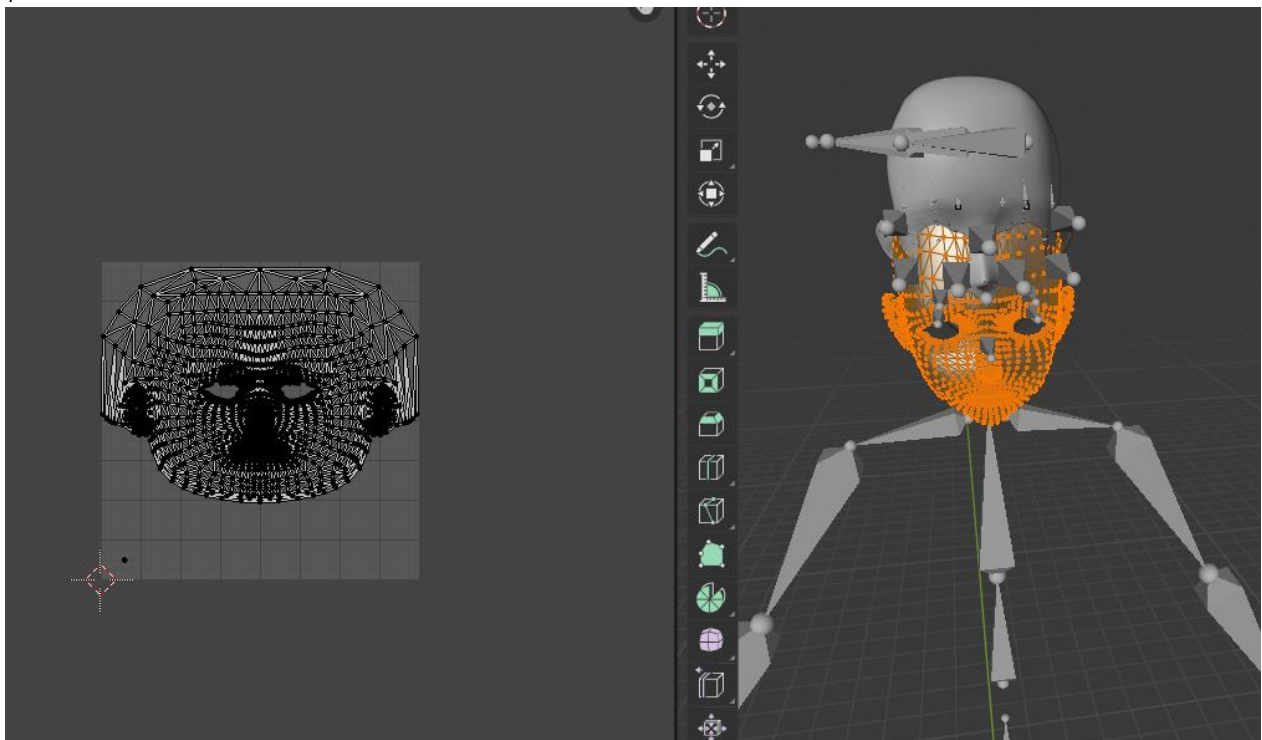
Base_0 will be transformed and have Sakura's face shape (note that the translation applied in Object mode was ignored) but with some of the necessary data (vertex groups) for export. We don't need Sakura's face anymore, we can either let it be or hide it (H) or delete it. Let's work on the base_0 mesh now.

2) Mesh transformation

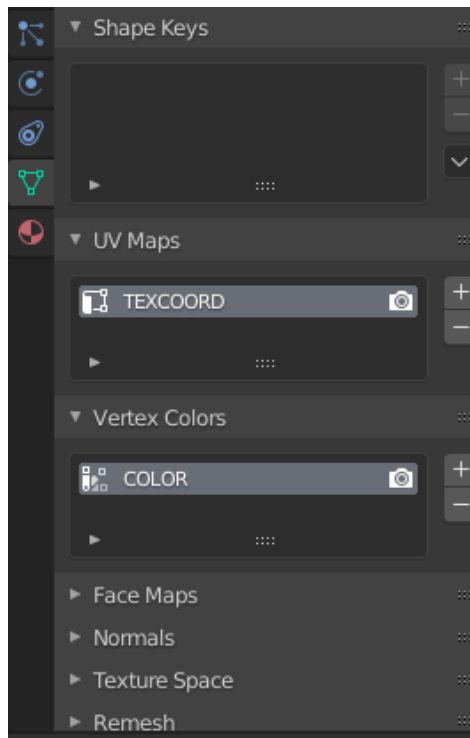
You may want to move, rotate, scale the whole mesh. Here for example Sakura's head is too low so we need to translate it a bit. **ALL THE OPERATIONS MUST BE DONE IN EDIT MODE, NOT OBJECT MODE.** You can do whatever edit you want, move vertices, create new ones etc, just remember to do all of them in edit mode when working on the whole mesh as we're working in model space.

3) UV maps, color channels

The UV maps of the mesh we want to put in the game are preserved. You can edit the coords however you want if needed.



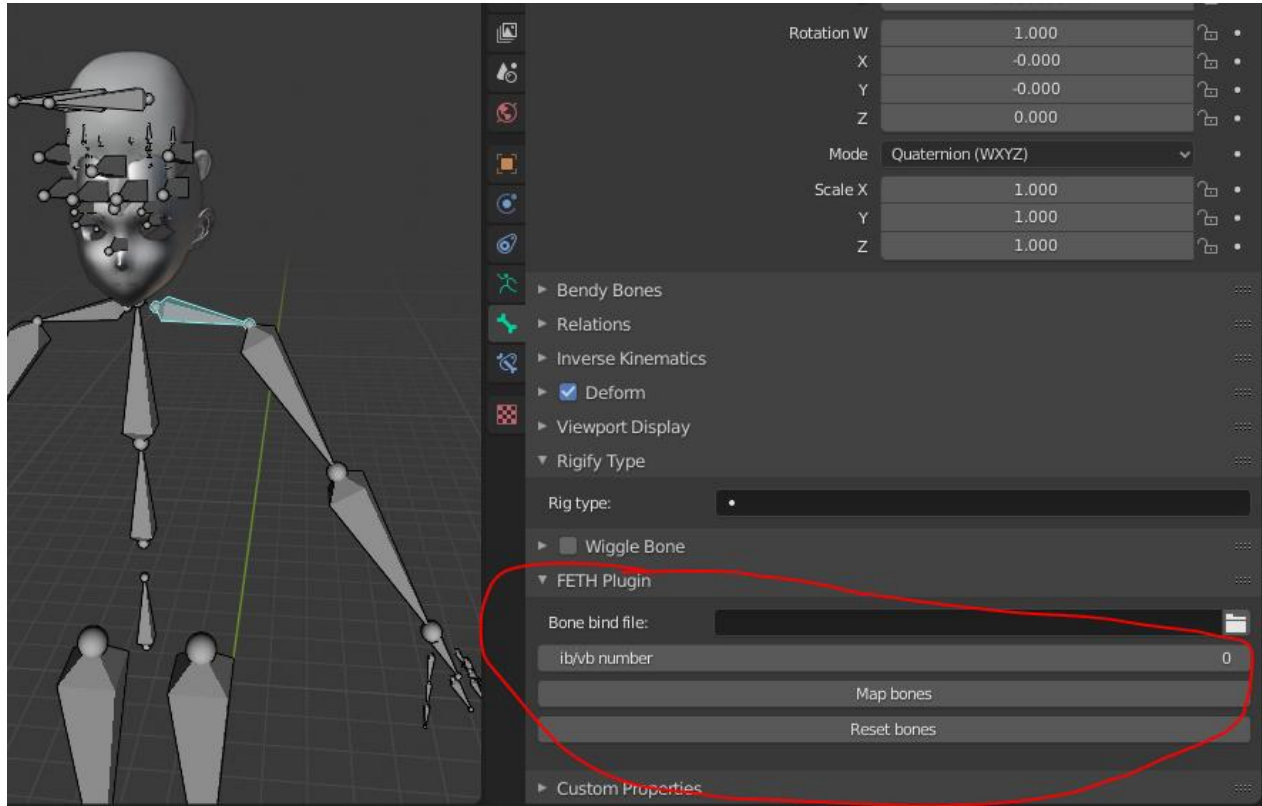
What you just need to do is rename it to “TEXCOORD” in the Object data properties panel, add a “COLOR” item in the Vertex Colors tab too.



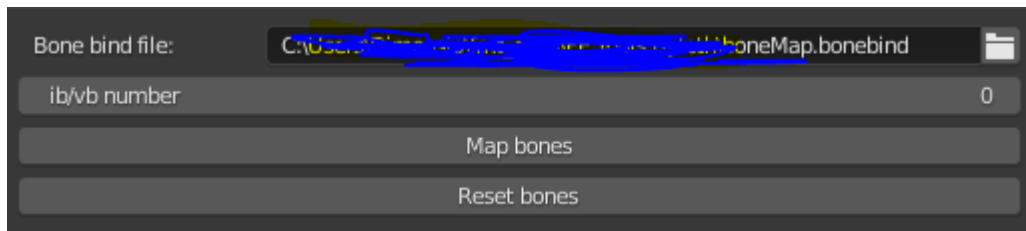
4) Rigging

Each submesh has a bone palette that you'll be able to use for rigging (the bones deforming the submesh). You won't be able to add new bones to that palette or modify the bones' position/rotation, as they are too many cross dependencies with other sections and unknown areas.

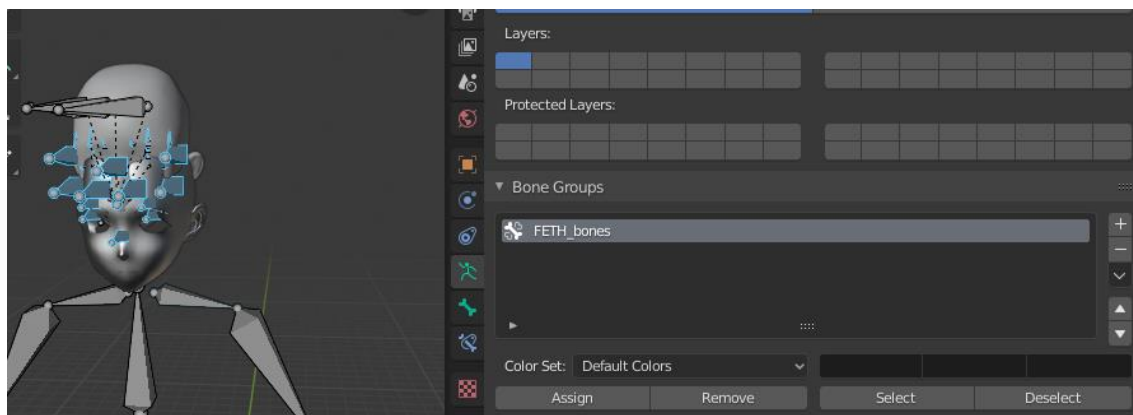
Let's get the bone palette for the 0.vb submesh. First select the armature, then go into pose mode. Select a random bone and then go to the Bone properties. Look for the FETH Plugin tab



Put the bonebind file as asked



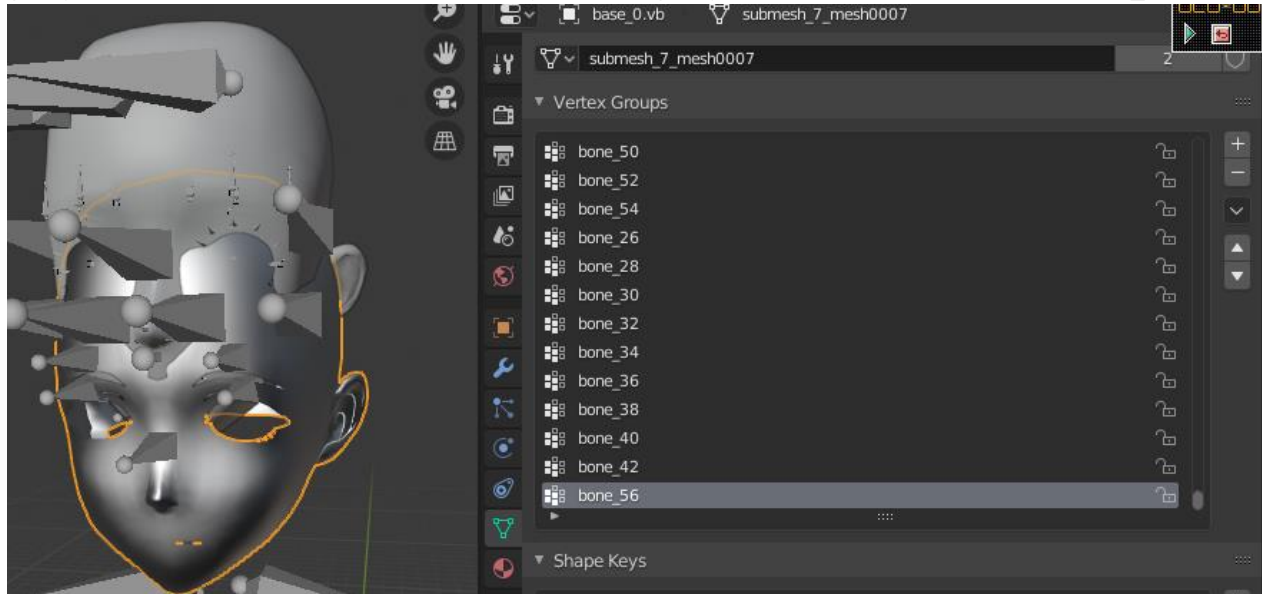
Then choose the ib/vb number. Here we're working on 0.vb so we won't change anything. Now just click on "Map bones".



(Note that I didn't bother translating Sakura's mesh for this example but you should align the mesh according to the skeleton obviously).

The bone palette will be selected and a new bone group with these will be created. I recommend to give them a special color or hide the others (CTRL +I then H). If you lost track on them just click on the bone Group and click on select or check their names: **the bones that you'll be able to use for rigging that mesh are named as numbers (0,3, etc) whereas the others will still have the "bone_x" names.**

Now it's time to rig the armature, select Sakura's head, select the armature and then choose "with empty groups". You'll notice that unnecessary vertex groups were created for the "bone_x" bones



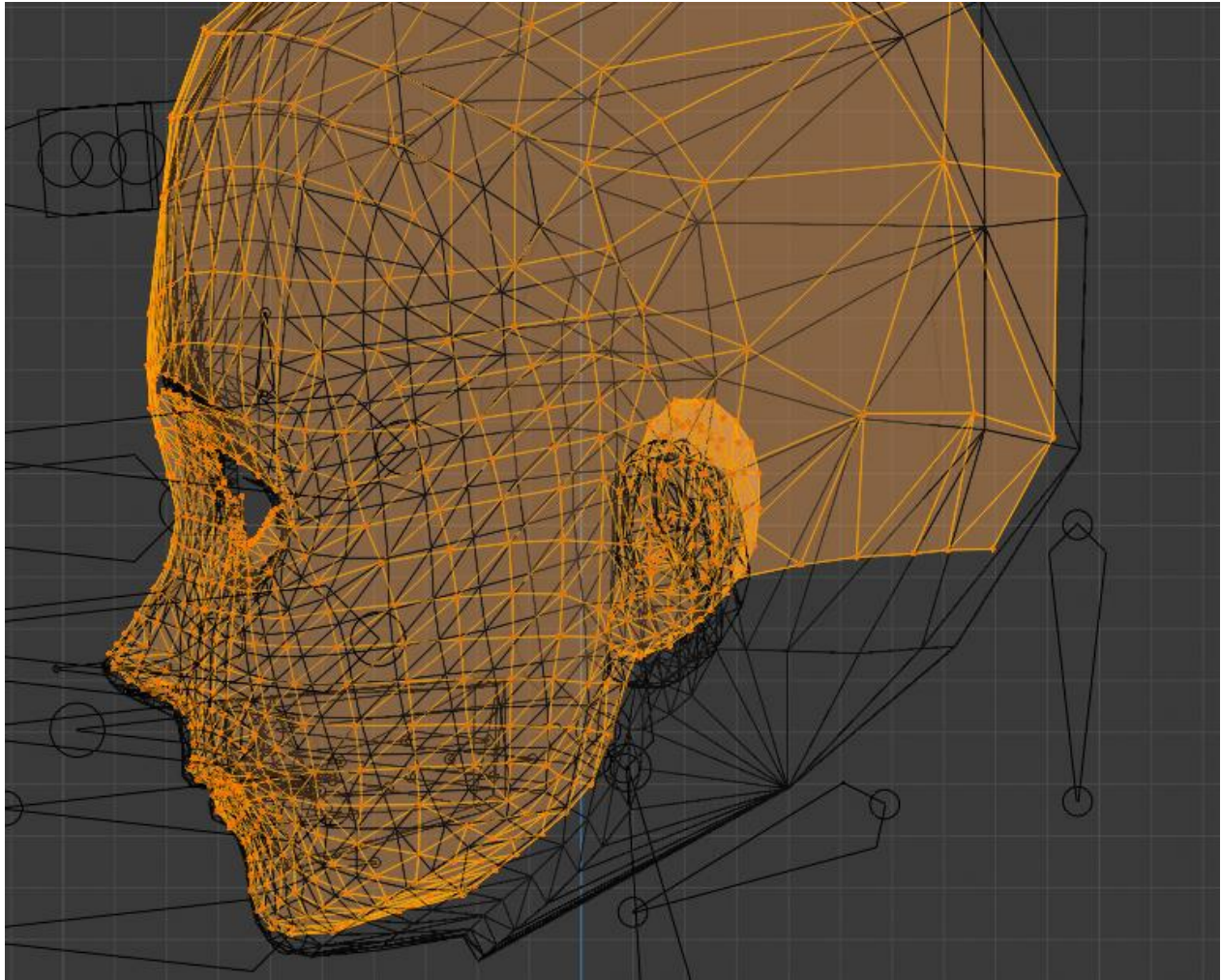
To get rid of them simply click on the arrow below the minus sign and choose "FETH remove extra vg". The mesh is now rigged to the bone palette and you can use weight painting or whatever you want to rig.

WHEN YOU'RE DONE WITH ONE MESH AND WANT TO RESET THE SKELETON, KEEP THE VB/IB INDEX THAT YOU USED AND CLICK ON "RESET BONES" AND THEN CHANGE THE VB/IB INDEX TO THE ONE OF THE NEW MESH, IT'LL COMPLETELY MESS THE MAPPING OTHERWISE AND THE FINAL RESULT WILL BE WRONG.

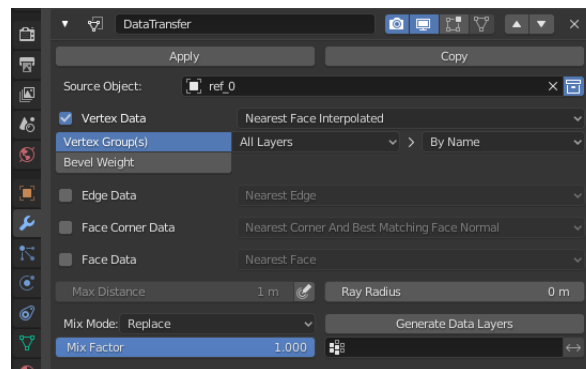
5) Weight transfer

Weight transfer is a fast way to transfer the weight paint from one mesh to another based on the distance between faces/vertices.

First align the base_0 mesh as best as you can with the ref_0 mesh (**IN EDIT MODE**)



Then add a data transfer modifier, choose ref_0 as a source object, choose vertex_groups and nearest face interpolated. Then click on “generate data layers” and apply the modifier, the weights will be transferred from one mesh to another.



If you have meshes with similar shapes like in my case, I’d recommend to first do the data transfer to get most of the work done and then adjust the details using weight painting and the armature to check if it deforms correctly.

III) Exporting

1) Getting the new vb/ib files

Once you're done with the model, simply choose File > export > 3d migoto raw buffer and then overwrite the file you imported (I imported 0.vb, I'll save my file over it as 0.vb).

2) Repacking all the vb/ib into g1m

Once you modified all the files you wanted, go back to your original folder.

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Then drag and drop "FE_Anette.g1m" on "3.exe". It will take all the ib/vb from the FE_Anette folder and repack them into a g1m.

At this point if you want to check you result you should import the g1m using my fmt_g1m Noesis script. When you're sure you're done just drag and drop FE_Anette.g1m on 4.exe. It will transform it into a g1m file that you'll be able to put in the game.

Questions:

Q: How do I replace cloth meshes?

A: You can't, these meshes are not like the others as they are deformed at runtime using the physics engine and some vertex shader so way more RE work would be necessary to replace them. You can preview them using my plugin but that's it. You can get rid of them by loading their vb in Blender (they'll appear as weird deformed shapes), deleting all their vertices except a single triangle and then overwriting the vb on export.

Q: How do I get rid of meshes?

A: Either make them really small or use the same triangle technique as above.

Credits :

-Eternity, for the g1mtools (2.exe and 3.exe)

-Ausgeek for the 3DMigoto plugin